

A Human-Centered Framework for Exploring and Steering LLM Internals: A Case Study of Bias in Essay Scoring

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Abstract

Large language models (LLMs) are used in high-stakes domains, yet mechanistic interpretability techniques are typically presented as isolated procedures rather than tools supporting human inquiry. We present a human-centered framework that reframes interpretability as a process of sensemaking, experimentation, and judgment through five stages: noticing value tensions, building mental models, interrogating causality, exploring controllability, and making normative judgments. While applicable to any undesirable model behavior, we demonstrate the framework through a case study of demographic bias in LLM-based essay scoring using the PERSUADE 2.0 corpus and Llama-3.1-8B-Instruct. Through linear probing, attribution patching, and activation steering, we show that practitioners can localize where bias emerges and manipulate its effects—achieving up to 83.3% reduction in score gaps while improving accuracy by up to 20.9%. We also uncover a dual mechanism of bias operating through separate pathways, highlighting limits of single-intervention approaches. Our contribution lies in articulating interpretability as an exploratory relationship between humans and models.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; *HCI design and evaluation methods*; • **Computing methodologies** → *Natural language processing*.

Keywords

Mechanistic interpretability; Bias mitigation; Large language models; Automated essay scoring; Human-centered AI

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1 Introduction

Large Language Models (LLMs) are increasingly deployed in high-stakes domains such as education, healthcare, and hiring, where they shape judgments, opportunities, and outcomes [10]. As these systems become embedded in consequential decision-making, questions about their internal behavior, including but not limited to fairness and bias, have become practical design challenges faced by practitioners building real systems.

Prior work has made substantial progress in documenting undesirable LLM behaviors through auditing methods, benchmarks, and algorithmic interventions [3, 8]. However, an important gap remains: how do practitioners actually come to understand where problematic behavior originates within a model, and how can they meaningfully explore whether and how to intervene?

In practice, undesirable behavior often first appears as a moment of surprise, when a system behaves in ways misaligned with intended values and prompts questions like "Why did it do that?" or "Can we change it?" These questions are not merely technical; they are interpretive, exploratory, and deeply human. Yet most interpretability tools are designed around algorithmic inspection rather than human inquiry [11].

In this paper, we argue that engaging with LLM internals should be understood as a human-centered process of sensemaking and decision-making. We propose a framework that reframes interpretability around the practitioner's experience: noticing problematic behavior, forming hypotheses, experimentally testing them, exploring controllability limits, and making value-laden design judgments. While applicable to investigating any undesirable model behavior, such as hallucination, safety violations, or miscalibration, we demonstrate the framework through a case study of demographic bias in essay scoring. From this process, we contribute:

- A five-stage human-centered framework for engaging with LLM internals, framing interpretability as an experiential workflow rather than isolated techniques.

- Empirical evidence demonstrating how practitioners can move from observing undesirable behavior to meaningful intervention, including 83.3% reduction in demographic score gaps and a corresponding 0.073 increase in correlation ($r: 0.350 \rightarrow 0.423$).
- Design implications for tools that support exploration, reflection, and responsible AI development.

2 Related Work

Our work bridges three research areas: behavior auditing in LLMs, mechanistic interpretability, and human-centered AI transparency.

Behavior auditing research has documented various undesirable patterns in LLM outputs across high-stakes domains [3, 8]. In educational contexts, studies have examined fairness in automated essay scoring systems [13, 19], with recent work showing that LLMs exhibit scoring biases related to demographic backgrounds [9]. However, this body of work primarily treats problematic behavior as an output-level property to be measured, offering limited insight into how practitioners might explore its origins within models.

Mechanistic interpretability provides powerful techniques for analyzing model internals, including linear probing [1, 2], attribution patching [14, 17], and activation steering [16, 20]. While these methods enable access to internal representations, they are typically presented as isolated technical procedures rather than tools supporting human inquiry and reflective judgment.

Human-centered AI transparency research emphasizes that interpretability is an experiential quality shaped by human interaction [6, 12]. Work on explainable AI has shown that effective tools should support sensemaking and mental model formation [11, 18]. We extend this tradition by examining how practitioners engage with LLM internals during system development—framing interpretability not as static inspection, but as an exploratory relationship involving experimentation and value-laden decision-making.

3 A Human-Centered Framework for Exploring Model Behavior

We conceptualize engagement with LLM internals not as a technical pipeline, but as a process of human sensemaking and judgment. We articulate a five-stage framework capturing how practitioners come to notice, interrogate, and shape model behavior (Figure 1). While we illustrate each stage through our bias case study, the framework generalizes to any scenario in which practitioners seek to understand and address undesirable model behavior.

Stage 1: Noticing Value Tensions. The process begins when practitioners encounter behavior that feels surprising or normatively ambiguous. This is not merely statistical anomaly detection, but reflective discomfort that motivates deeper inquiry—driven by concerns about what the system *ought* to be doing [7].

Stage 2: Building Mental Models. Practitioners construct tentative mental models of the system’s internal reasoning. Techniques like linear probing [1] scaffold this cognitive process, supporting questions such as: Is specific information explicitly represented? Which layers are most sensitive? Probing supports sensemaking rather than merely producing measurements [18].

Stage 3: Interrogating Causality. As mental models become concrete, questions shift from correlation to causation: “If this component matters, what happens if we intervene?” Attribution patching [17] enables experimental dialogue with the model—practitioners actively test hypotheses through intervention rather than passive inspection.

Stage 4: Exploring Controllability. Steering experiments [16] introduce new questions: not just where behavior originates, but how controllable it is. Practitioners probe the limits, asymmetries, and directional sensitivities of model behavior—understanding what kinds of influence the system permits or resists.

Stage 5: Making Normative Judgments. The final stage moves beyond technical manipulation into design judgment. Practitioners must decide whether and how interventions should be applied, weighing trade-offs and considering unintended consequences. Addressing undesirable behavior becomes a value-laden design decision, not purely technical optimization [15].

4 Case Study: Exploring Bias in an LLM-Based Scoring System

We illustrate our framework through a case study of demographic bias in essay scoring. We used the PERSUADE 2.0 corpus [4] ($n=3,000$ essays; holistic scores on a 1–6 scale assigned by human raters) and Llama-3.1-8B-Instruct [5], comparing base and QLoRA-finetuned model variants. Bias was measured by comparing scores generated with and without demographic information in the prompt. For linear probing, we extracted activation vectors at the final token position across layers 5, 10, 15, 20, 25, 30 and trained logistic regression classifiers to predict English Language Learner (ELL) status (5-fold cross-validation, $n=200$ per group). Attribution patching measured causal effects as the L2 norm of activation differences between demographic conditions ($n=50$). Steering vectors were computed as the mean activation difference between demographic conditions and added to activations during inference at Layer 25.

4.1 Noticing Value Tensions

Providing ELL status in prompts increased predicted scores by +0.44 points, even with identical essay content. This surfaced a normative tension rather than a mere statistical anomaly: Is the model exhibiting thoughtful accommodation or inappropriate overcompensation? Such questions motivated deeper exploration.

4.2 Building Mental Models

Linear probing revealed that ELL status was encoded across transformer layers with 76–80% classification accuracy, strongest in layers 25–30. Probing functioned not as definitive diagnosis, but as hypothesis formation: demographic information appeared meaningfully represented in later processing stages.

4.3 Interrogating Causality

Attribution patching, measured as the L2 norm of activation differences between control and demographic-informed prompts ($n=50$), showed a clear gradient—interventions in later layers produced substantially stronger effects, with Layer 30 showing approximately

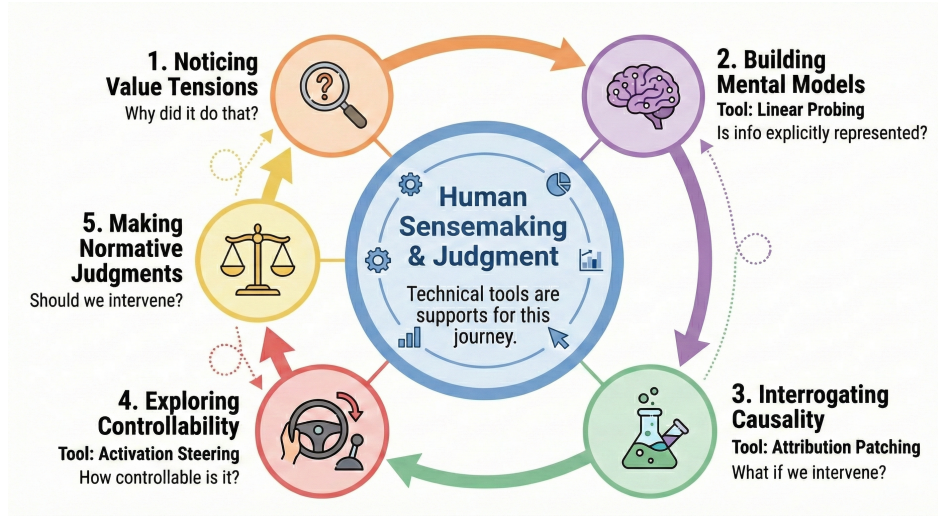


Figure 1: A five-stage human-centered framework for exploring LLM internals.

5× the causal influence of Layer 15. This shifted our understanding: bias appeared concentrated in higher-level representational processing rather than diffusely distributed.

4.4 Exploring Controllability

Steering experiments at Layer 25 with scaling factor α ranging from 0.5 to 2.5 (averaged across two random seeds) revealed a 4:1 asymmetry: positive steering (toward ELL representations) produced large score shifts, while negative steering yielded minimal effects. Fine-tuning reduced this asymmetry to 1.3:1, indicating that training reshapes the internal controllability landscape. This asymmetry suggests Non-ELL text functions as an implicit “default,” with ELL characteristics representing penalized deviations.

4.5 Making Normative Judgments

Applying negative steering ($\alpha=1.5$) at Layer 25 to ELL essays scored without demographic prompts reduced the ELL–Non-ELL score gap by 83.3% (base model) and 70.8% (fine-tuned). Crucially, this intervention did not come at the cost of performance; instead, it improved scoring accuracy. The correlation with human scores increased from 0.350 to 0.423 (+0.073) in the base model and from 0.393 to 0.443 (+0.050) in the fine-tuned model.

This “win-win” outcome shifts the normative interpretation of the model’s behavior. The initial score gap was not a “thoughtful accommodation” but rather a form of inferential noise caused by implicit demographic bias. By manipulating internal activations to suppress this bias, the model’s judgment became more aligned with human evaluation, suggesting that internal bias manipulation can serve as a path to both fairness and functional optimization.

However, we discovered a dual mechanism: while steering effectively addressed implicit bias encoded in text representations, it could not counteract explicit compensation triggered by demographic keywords in prompts—these pathways operate independently. When steering was applied to essays scored with explicit ELL information in the prompt, the compensation effect was not

reduced but slightly increased, yielding a negative cancellation rate (−31.8% in the base model). Determining appropriate intervention thus required interpretive judgment about trade-offs, not mere optimization.

5 Design Implications

Our findings suggest that interpretability should be conceived as an experiential practice, not merely a technical capability. Although derived from a bias mitigation case study, we argue these implications extend to any practitioner engagement with model internals. We outline implications for human-centered tool design.

Support Exploratory Practice, Not Static Inspection. Current tools emphasize visualization and inspection (heatmaps, attribution plots). Our case study shows practitioners’ needs evolve—from noticing behavior, to hypothesis formation, to causal testing, to evaluating trade-offs. Tools should support iterative exploration, enabling fluid movement between observation, experimentation, and reflection [12].

Make Controllability Legible. Steering experiments revealed that not all behaviors are equally controllable—some directions produced large effects while others yielded weak responses. Fine-tuning altered this landscape. Future interfaces should visualize where intervention is likely effective, surface asymmetries, and warn when manipulations produce unstable outcomes. Making controllability visible helps practitioners develop realistic expectations about feasible changes.

Frame Behavior Correction as Design Judgment. While addressing undesirable model behavior is often framed as a trade-off between competing objectives, our findings suggest a different paradigm: undesirable behavior as a performance bottleneck. In our case study, steering not only reduced demographic gaps but also acted as an “accuracy booster” by removing confounding internal representations.

Future tools should help practitioners identify whether a specific behavior pattern is a “functional error” (which, when removed,

improves accuracy) or a “value-based preference” (which might require a trade-off). Supporting this distinction is critical for making informed normative judgments. Systems should visualize how interventions affect not only the target behavior but also the overall reliability of the system’s core task.

Design for Interpretability as Relationship. Across our findings, interpretability was experienced not as accessing “truth” about a model, but as developing a relationship through interaction and experimentation. Rather than treating interpretability as a model property, we might design for interpretability as a relationship—one where tools mediate how practitioners learn about, test, and take responsibility for system behavior.

6 Contributions and Limitations

This work reframes engagement with LLM internals from a purely technical exercise to a human-centered process of exploration, sensemaking, and judgment. Our primary contribution is not new interpretability algorithms, but a conceptual shift: existing techniques such as probing, attribution patching, and steering can function as tools for human inquiry rather than isolated procedures. By framing internal analysis as an exploratory relationship between humans and models, we argue that responsible AI development requires tools that support reflection, experimentation, and normative decision-making.

Through our bias case study, we demonstrate that these internal interventions can achieve a dual benefit by reducing demographic score gaps by up to 83.3% while simultaneously improving scoring accuracy, measured as correlation with human scores, by as much as 20.9% relative improvement. The discovery of a dual mechanism, in which steering addresses implicit bias but not explicit compensation, illustrates how the framework’s iterative nature helps practitioners uncover nuances that single-technique approaches may miss.

We hope this work encourages the HCI community to consider not only how to make AI systems more transparent, but also how to better support the people who must interpret, interrogate, and take responsibility for their behavior.

We acknowledge several important limitations. First, our case study focuses on a single domain (essay scoring), a single model family (Llama 3.1), and a single demographic dimension (ELL status). The framework’s generalizability to other domains, such as healthcare or hiring, other model architectures, such as GPT or Claude, and other types of undesirable behavior, such as hallucination or safety violations, requires further investigation. Second, interventions were performed by technically skilled researchers rather than through a realized user-facing interface; the framework has not yet been validated with end-user practitioners. We plan to conduct a formative study with 10–15 ML practitioners using think-aloud protocols and semi-structured interviews to examine how they navigate the five stages, where they encounter difficulties, and what tool support would be most valuable. Third, the dual mechanism we identified, in which steering addresses implicit bias but not explicit compensation, requires further investigation across different models and contexts. We do not claim generalizability to all practitioners or all model behaviors; rather, we offer a conceptual foundation for future empirical research.

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